

● RAG Learning

4.5 Pythagoras & Trigonometry

AQA GCSE Mathematics 8300 — Geometry and measures — spec refs G20–G23 — Papers 1 (non-calculator), 2 and 3 (calculator)

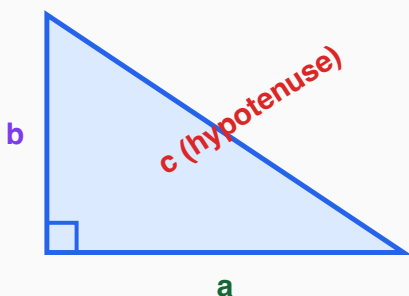
This topic is about finding missing lengths and angles in triangles. Foundation students use Pythagoras' theorem and the three trigonometric ratios in right-angled triangles in two dimensions, and learn the exact values of sin, cos and tan for the special angles. Higher students extend this to three-dimensional problems and to non-right-angled triangles with the sine rule, the cosine rule and the formula $\frac{1}{2}ab \sin C$ for area.

4.5.1 Pythagoras' Theorem

Pythagoras' theorem: in any **right-angled triangle**, the square of the **hypotenuse** (the longest side, opposite the right angle) equals the sum of the squares of the other two sides.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse. The specification requires you to know this formula.



Finding the hypotenuse (square, *add*, square root):

$$c = \sqrt{a^2 + b^2}$$

Finding a shorter side (square, *subtract*, square root):

$$a = \sqrt{c^2 - b^2}$$

Worked Example: A right-angled triangle has shorter sides 9 cm and 12 cm. Find the hypotenuse.

$$c^2 = 9^2 + 12^2 = 81 + 144 = 225$$

$$c = \sqrt{225} = \mathbf{15 \text{ cm}}$$

Worked Example: A 5 m ladder leans against a vertical wall with its foot 1.4 m from the wall. How far up the wall does it reach?

The ladder is the hypotenuse, so subtract: $h^2 = 5^2 - 1.4^2 = 25 - 1.96 = 23.04$

$$h = \sqrt{23.04} = \mathbf{4.8 \text{ m}}$$

Pythagorean Triples

Whole-number solutions of $a^2 + b^2 = c^2$ are worth recognising — they often appear on the non-calculator paper.

Triple	Some multiples
3, 4, 5	6, 8, 10 9, 12, 15 30, 40, 50
5, 12, 13	10, 24, 26
8, 15, 17	16, 30, 34
7, 24, 25	14, 48, 50

Is the Triangle Right-Angled?

The **converse** of Pythagoras: if the sides satisfy $a^2 + b^2 = c^2$ (with c the longest side), the triangle has a right angle opposite c . If $a^2 + b^2 \neq c^2$ it is not right-angled.

Worked Example: Show that a triangle with sides 20 cm, 21 cm and 29 cm is right-angled.

$$20^2 + 21^2 = 400 + 441 = 841 \text{ and } 29^2 = 841.$$

These are equal, so the triangle **is right-angled** (the right angle is opposite the 29 cm side).

Distance Between Two Points

Draw a right-angled triangle using the horizontal and vertical differences between the coordinates:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Worked Example: Find the distance between $A(-3, 4)$ and $B(5, -2)$.

Horizontal difference = $5 - (-3) = 8$; vertical difference = $-2 - 4 = -6$

$$d = \sqrt{8^2 + 6^2} = \sqrt{100} = \mathbf{10 \text{ units}}$$

Isosceles and Equilateral Triangles

The line of symmetry of an isosceles triangle cuts it into two congruent right-angled triangles, so Pythagoras gives the perpendicular height.

Worked Example: An isosceles triangle has sides 13 cm, 13 cm and base 10 cm. Find its area.

$$\text{Half the base is 5 cm. Height} = \sqrt{13^2 - 5^2} = \sqrt{144} = 12 \text{ cm}$$

$$\text{Area} = \frac{1}{2} \times 10 \times 12 = \mathbf{60 \text{ cm}^2}$$

Exam Tip: Always identify the hypotenuse first. A very common error is adding the squares when you need a shorter side. Check your answer: the hypotenuse must be the longest side.

4.5.2 Trigonometric Ratios (SOHCAHTOA)

In a right-angled triangle, label the sides relative to the angle θ you are using:

- **Hypotenuse (H)** — the longest side, opposite the right angle
- **Opposite (O)** — the side opposite θ
- **Adjacent (A)** — the side next to θ that is not the hypotenuse

The trigonometric ratios are fixed for each angle because all right-angled triangles with the same angle θ are **similar**.

SOH

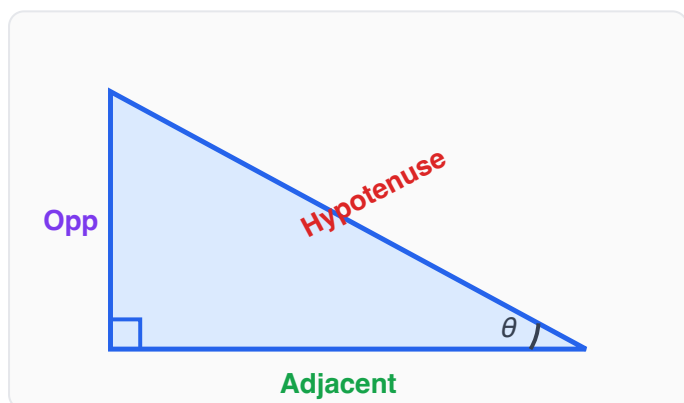
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

CAH

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

TOA

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



Method

1. Label the sides O, A, H relative to the angle.
2. Tick the two sides involved (the one you know and the one you want).
3. Choose the ratio that uses those two sides.
4. Substitute and rearrange. To find an **angle**, use the inverse function: $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$.

Worked Example: Missing side (on top): In triangle ABC, angle B = 90° , angle A = 52° and AC = 15 cm. Find BC.

BC is opposite 52° and AC is the hypotenuse, so use SOH: $\sin 52^\circ = \frac{BC}{15}$
 $BC = 15 \sin 52^\circ = 11.820 \dots \approx \mathbf{11.8 \text{ cm}}$

Worked Example: Missing side (underneath): The side adjacent to 25° is x and the opposite side is 80 m. Find x .

$$\text{TOA: } \tan 25^\circ = \frac{80}{x}, \text{ so } x = \frac{80}{\tan 25^\circ} = 171.56 \dots \approx \mathbf{172 \text{ m}}$$

Worked Example: Missing angle: The adjacent side is 6 cm and the hypotenuse is 11 cm. Find θ .

$$\text{CAH: } \cos \theta = \frac{6}{11}, \text{ so } \theta = \cos^{-1}\left(\frac{6}{11}\right) = 56.94 \dots \approx \mathbf{56.9^\circ}$$

Angles of Elevation and Depression

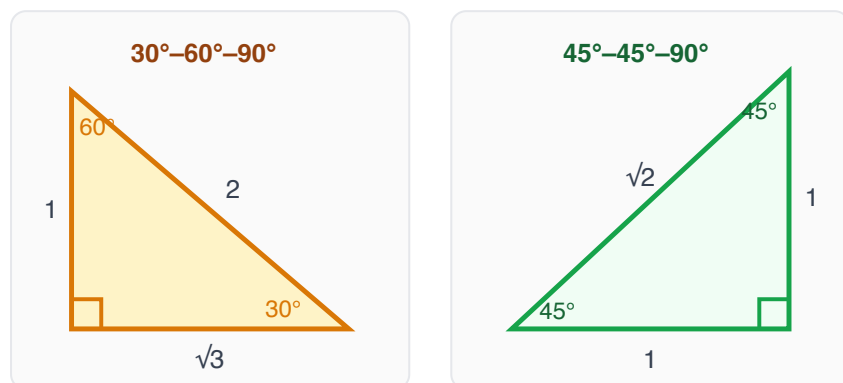
Both are measured **from the horizontal**. The angle of elevation looks up; the angle of depression looks down. By alternate angles, the angle of depression from A to B equals the angle of elevation from B to A.



Exam Tip: Make sure your calculator is in **degrees** mode (a small D on the display). Keep the full calculator value for later steps and only round your final answer.

4.5.3 Exact Trigonometric Values

You must know these values without a calculator (spec ref G21). They come from two special triangles.



The 30°–60°–90° triangle is half of an equilateral triangle of side 2: the height is $\sqrt{2^2 - 1^2} = \sqrt{3}$. The 45° triangle is a right-angled isosceles triangle with shorter sides 1, so the hypotenuse is $\sqrt{2}$.

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	not defined

Note that $\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$; either form is accepted. A useful pattern: the sin row reads $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$, and the cos row is the same list backwards.

Worked Example: Without a calculator, find the exact length of the side opposite 60° in a right-angled triangle with hypotenuse 8 cm.

$$\text{opposite} = 8 \sin 60^\circ = 8 \times \frac{\sqrt{3}}{2} = 4\sqrt{3} \text{ cm}$$

Worked Example: Work out $\sin 30^\circ + \cos 60^\circ + \tan 45^\circ$.

$$= \frac{1}{2} + \frac{1}{2} + 1 = 2$$

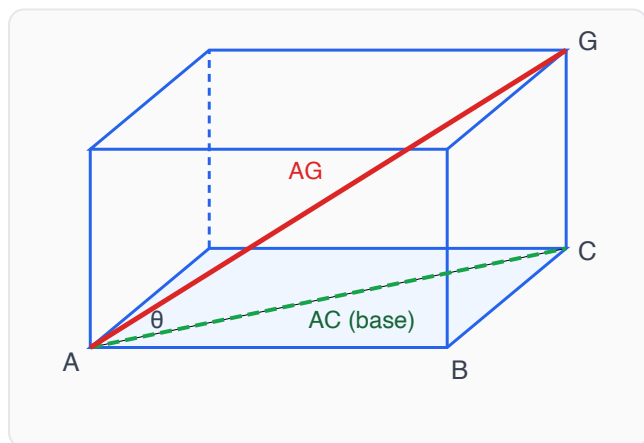
Exam Tip: On Paper 1 (non-calculator), any trigonometry question will use these special angles. "Give an exact answer" means leave surds such as $4\sqrt{3}$ — do not convert to decimals.

4.5.4 Pythagoras and Trigonometry in 3D Higher

Find a right-angled triangle that lies in a flat plane inside the 3D shape, draw it separately, then use Pythagoras or SOHCAHTOA. You often need two triangles: one in the base, then one standing up.

Space diagonal of a cuboid with edges a, b, c :

$$d = \sqrt{a^2 + b^2 + c^2}$$



Worked Example: Cuboid ABCDEFGH has $AB = 5$ cm, $BC = 4$ cm and $CG = 3$ cm. Find the length AG and the angle between AG and the base $ABCD$.

Base: $AC^2 = 5^2 + 4^2 = 41$

Triangle ACG (right angle at C): $AG = \sqrt{41 + 3^2} = \sqrt{50} = 7.07 \dots \approx \mathbf{7.1 \text{ cm}}$

$$\tan \theta = \frac{CG}{AC} = \frac{3}{\sqrt{41}}, \text{ so } \theta = \mathbf{25.1^\circ}$$

Angle Between a Line and a Plane

Drop a perpendicular from the top of the line down to the plane. The angle you want is between the line and the line joining its foot to where the perpendicular lands (its **projection**).

Worked Example: A pyramid has a square base of side 6 cm. The apex is directly above the centre of the base and each sloping edge is 9 cm. Find the height and the angle a sloping edge makes with the base.

$$\text{Half of the base diagonal} = \frac{1}{2}\sqrt{6^2 + 6^2} = 3\sqrt{2} = 4.24 \dots \text{ cm}$$

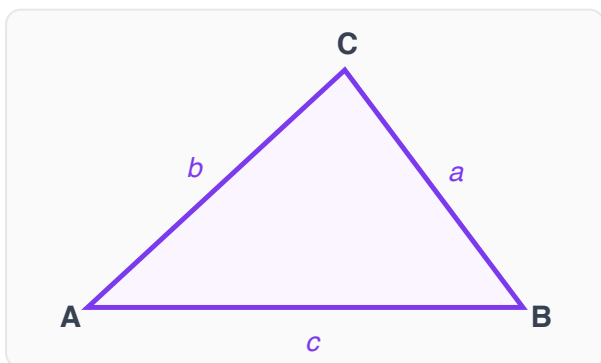
$$\text{Height} = \sqrt{9^2 - (3\sqrt{2})^2} = \sqrt{81 - 18} = \sqrt{63} \approx \mathbf{7.94 \text{ cm}}$$

$$\cos \theta = \frac{3\sqrt{2}}{9}, \text{ so } \theta \approx \mathbf{61.9^\circ}$$

Exam Tip: Sketch each 2D triangle separately and mark the right angle. In a pyramid, the apex is above the centre of the base, so you need *half* the diagonal, not the whole diagonal.

4.5.5 The Sine Rule Higher

For triangles **without** a right angle, label the vertices A, B, C and the sides opposite them a, b, c .



Sine rule (the specification requires you to know it):

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Use it when you know a side **and its opposite angle** (a matching pair), plus one other side or angle.

Worked Example: In triangle ABC, $A = 48^\circ$, $B = 73^\circ$ and $a = 12$ cm. Find b .

$$\frac{b}{\sin 73^\circ} = \frac{12}{\sin 48^\circ} \Rightarrow b = \frac{12 \sin 73^\circ}{\sin 48^\circ} = 15.44 \dots \approx \mathbf{15.4 \text{ cm}}$$

Worked Example: In triangle ABC, $a = 10$ cm, $b = 13$ cm and $B = 70^\circ$. Find angle A .

$$\frac{\sin A}{10} = \frac{\sin 70^\circ}{13} \Rightarrow \sin A = \frac{10 \sin 70^\circ}{13} = 0.7228 \dots$$

$$A = \sin^{-1}(0.7228 \dots) = \mathbf{46.3^\circ} \text{ (the alternative } 133.7^\circ \text{ is impossible because } 133.7^\circ + 70^\circ > 180^\circ \text{)}$$

The Ambiguous Case

Because $\sin \theta = \sin(180^\circ - \theta)$, an equation such as $\sin B = 0.8264$ has two solutions between 0° and 180° : $B = 55.7^\circ$ or $B = 180^\circ - 55.7^\circ = 124.3^\circ$. When you are given two sides and an angle that is *not* between them, check whether both fit with the angle sum. For $a = 7$, $b = 9$, $A = 40^\circ$: $40^\circ + 124.3^\circ < 180^\circ$, so **two** different triangles are possible.

Exam Tip: Put the unknown on top: use $\frac{a}{\sin A}$ to find a side and $\frac{\sin A}{a}$ to find an angle. You only ever need two of the three fractions.

4.5.6 The Cosine Rule Higher

Cosine rule (the specification requires you to know it):

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Rearranged to find an angle:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Use it when you know **two sides and the included angle** (SAS) or **all three sides** (SSS).

Worked Example: In triangle ABC, $AB = 11$ cm, $AC = 8$ cm and angle $BAC = 37^\circ$. Find BC .

$$BC^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 37^\circ = 185 - 140.55 \dots = 44.44 \dots$$

$$BC = \sqrt{44.44 \dots} = 6.666 \dots \approx \mathbf{6.67 \text{ cm}}$$

Worked Example: A triangle has sides 7 cm, 9 cm and 12 cm. Find the largest angle.

The largest angle is opposite the longest side (12 cm):

$$\cos \theta = \frac{7^2 + 9^2 - 12^2}{2 \times 7 \times 9} = \frac{-14}{126} = -0.111 \dots$$

$$\theta = \cos^{-1}(-0.111 \dots) = \mathbf{96.4^\circ}$$
 — a negative cosine means the angle is obtuse.

Exam Tip: Work out $b^2 + c^2$ and $2bc \cos A$ separately before subtracting — typing it all at once often goes wrong. Never use the cosine rule on a right-angled triangle; Pythagoras is quicker.

4.5.7 Area of a Triangle: $\frac{1}{2}ab \sin C$ Higher

For any triangle with two sides a and b and the **included** angle C between them:

$$\text{Area} = \frac{1}{2}ab \sin C$$

Worked Example: A triangle has sides 7 cm and 12 cm with an included angle of 50° . Find its area.

$$\text{Area} = \frac{1}{2} \times 7 \times 12 \times \sin 50^\circ = 42 \sin 50^\circ = 32.17 \dots \approx \mathbf{32.2 \text{ cm}^2}$$

Worked Example: A triangle has area 30 cm^2 and two sides 8 cm and 10 cm with an acute angle between them. Find that angle.

$$30 = \frac{1}{2} \times 8 \times 10 \times \sin C = 40 \sin C \Rightarrow \sin C = 0.75$$

$$C = \sin^{-1}(0.75) = \mathbf{48.6^\circ}$$

Worked Example: Multi-step bearings problem. A ship sails 12 km from P on a bearing of 040° to Q, then 9 km on a bearing of 110° to R. Find PR and the bearing of R from P.

The back bearing of P from Q is 220° , so angle $PQR = 220^\circ - 110^\circ = 110^\circ$.

$$PR^2 = 12^2 + 9^2 - 2 \times 12 \times 9 \cos 110^\circ = 298.87 \dots, \text{ so } PR \approx \mathbf{17.3 \text{ km}}$$

$$\sin QPR = \frac{9 \sin 110^\circ}{17.28 \dots} = 0.4892 \dots \Rightarrow QPR = 29.3^\circ$$

$$\text{Bearing of R from P} = 040^\circ + 29.3^\circ = \mathbf{069.3^\circ}$$

Choosing the Right Method

Information given	Want	Use
Right angle + two sides	Third side	Pythagoras
Right angle + one side + one angle (or two sides)	Side or angle	SOHCAHTOA

A side with its opposite angle + one more side/angle	Side or angle	Sine rule Higher
Two sides + included angle	Third side	Cosine rule Higher
Three sides	An angle	Cosine rule Higher
Two sides + included angle	Area	$\frac{1}{2}ab \sin C$ Higher

Exam Tip: The specification says you must *know* every formula in this topic — learn Pythagoras, SOHCAHTOA, the exact values, the sine rule, the cosine rule and $\frac{1}{2}ab \sin C$.